Dr. Gyu Seon Kim

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- Personal Website: https://gyuseonkim.github.io/
- Google Scholar: https://scholar.google.com/citations?user=53OQU2AAAAAJ
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- DBLP: https://dblp.org/pid/333/1068.html
- DBPia (Korean): https://www.dbpia.co.kr/author/authorDetail?ancId=5405761
- GitHub: https://github.com/GSKIMKOR
- IEEE Member: https://ieeexplore.ieee.org/author/37089737473

Research Interests

- **Autonomous Aerospace Systems:** Control and Communication for Autonomous Aerospace Systems (*e.g.*, Aircraft, Drone, UAV, UAM, Fixed Wing, Rotary Wing, Rotorcraft, Launch Vehicles, and Spacecraft)
- **Physical AI for Reinforcement-Learning-based Flight Control:** Developing reinforcement-learning-based flight controllers and enabling their closed-loop sim-to-real deployment on fighter aircraft and other real-world aerial platforms.
- **AI for Defense and Military Applications:** Intelligent Control and Autonomous Decision-Making for Defense and Aerospace Systems (*e.g.*, Fighter Aircraft, Military Drone, Anti Drone System, Missile, Decoy, and Torpedo)
- **MARL-based Autonomous Control:** Multi-Agent Reinforcement Learning for Cooperative and Networked Aerial Vehicle Systems
- **RL-based Satellite Communication Systems:** Reinforcement Learning-based Communication and Networking for Satellite Systems (*e.g.*, LEO Satellite Routing, Cube Satellite Design, Satellite Constellations, and Space-Air-Ground Integrated Networks)
- **Quantum AI:** Quantum Reinforcement Learning for Aerospace Control and Communication Systems

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Educational Background

- **Korea University – College of Engineering**, Seoul, Republic of Korea
  – Ph.D. in **Electrical and Computer Engineering (ECE)**
- **Inha University – College of Engineering**, Incheon, Republic of Korea
  – B.S. in **Aerospace Engineering**

Academic Background

- **Korea University – Research Institute of Information and Communication Technology**, Seoul, Republic of Korea
  – *Postdoctoral Researcher*
- **University of Houston – College of Engineering (ECE)**, Houston, TX, USA
  – *Visiting Scholar*, Wireless Networking, Signal Processing and Security Laboratory (Prof. Zhu Han)
- **Korea University – College of Engineering (ECE)**, Seoul, Republic of Korea
  – *Research Assistant*, Artificial Intelligence and Mobility Laboratory (Advisor: Prof. Joongheon Kim)

Industry Background

- **RealtimeVisual – R&D Center**, Seoul, Republic of Korea
  – *Aerospace AI Research Engineer (Military Service Exception)*

Awards and Honors

- **IEIE Best Paper Award** (11/2025)
  – *Title: Lightweight Control Modeling via Quantum Reinforcement Learning*
- **IPESK Next Generation Engineering Researcher Award** (10/2025)
  – *Funded by an Institute for Promotion of Engineering and Science of Korea*
- **IEEE Vehicular Technology Society Student Scholarship Award** (10/2025)
  – *Funded by an IEEE VTS Endowment Fund and partially funded by a VTS Award Fund, and administered by the IEEE Foundation*
- **KICS Best Paper Award** (11/2024)
  – *Title: Reinforcement Learning-Based Counter Measure Tactics to Avoid Torpedo Threats*
- **IEEE Seoul Section Student Paper Contest, Best Paper Award** (12/2023)
  – *Title: Aircraft Taxi Routing using Reinforcement Learning at Hartsfield Jackson Atlanta International Airport*
- **KICS Best Paper Award** (02/2023)
  – *Title: Self-Learning-Based Hybrid MAC for Military UAV Networks*

R&D Projects

- **18. Development of an Air-to-Air Combat Software Verification Environment**
  Funded by *ADD (Agency for Defense Development)*
- **17. KE-Anduril AI Pilot Project: Development of AI Algorithms and Simulation Frameworks for Mission Autonomy**
  Funded by *Korean Air*
- **16. Discovering Quantum-Advantage Cases for Industry Adoption**
  Funded by *KTL (Korea Testing Laboratory) and the Ministry of Trade, Industry and Resources (MOTIR), Republic of Korea*
- **15. LEO Satellite Routing Research using Large Language Model and Reinforcement Learning**
  Funded by *ETRI (Electronics and Telecommunications Research Institute)*
- **14. Quantum Reinforcement Learning for Satellite Backhaul Routing in Disaster Networks**
  Funded by *ETRI (Electronics and Telecommunications Research Institute)*
- **13. Research on Quantum Multi-Agent Reinforcement Learning Stability (Research on Multi-Agent Reinforcement Learning Exploration, Communication, Training Strategy)**
  Funded by *ETRI (Electronics and Telecommunications Research Institute)*
- **12. Net-Zero CAFE (Connectivity and Autonomy for Future Ecosystem) Research Center, ITRC**
  Funded by *IITP (Institute of ICT Promotion)*
- **11. Quantum AI Empowered Second-Life Platform Technology**
  Funded by *IITP (Institute of ICT Promotion), Software Technology Advanced Research*
- **10. Development of a 20-Ton Excavator and Application Services using the SDM (Software-Defined-Machine) Platform**
  Funded by *KOCETI (Korea Construction Equipment Technology Institute) and Hyundai*
- **9. Research on Learning-based Swarm Mission Planning Algorithms**
  Funded by *LIG Defense & Aerospace and ADD (Agency for Defense Development)*
- **8. AI Bots Collaborative Platform and Self-Organizing Artificial Intelligence Technology Development**
  Funded by *NRF (National Research Foundation of Korea)*
- **7. Quantum Hyper-Driving: Quantum Hyper-Connected and Hyper-Sensing Autonomous Mobility**
  Funded by *NRF (National Research Foundation of Korea)*
- **6. Distributed Learning System Design and Implementation for Clinical Applications**
  Funded by *Cipherome*
- **5. Routing Algorithms for LEO Satellite Networks**
  Funded by *Solvit System*
- **4. Optimal Positioning Algorithms for Wide-Area Relaying Networks**
  Funded by *Solvit System*
- **3. Advancement Technology Development for Submarine Target Identification and Engagement Support Intelligence**
  Funded by *LIG Defense & Aerospace*
- **2. Advancement Technology Development for Torpedo Deception Strategies in Submarines**
  Funded by *LIG Defense & Aerospace*
- **1.Nano UAV Intelligence Systems Research Lab (NUiSRL)**
  Funded by *ADD (Agency for Defense Development)*

Publications (International)

Journals (SCI) and Magazines

◄ Review ►

[TIV.*review*] [29] **Gyu Seon Kim**, Hoyeong Lee, Soohyun Park, Soyi Jung, Zhu Han, and Joongheon Kim, "LLM-Guided Reinforcement Learning for Carbon-Neutral Intelligent Vehicle Control," *IEEE Transactions on Intelligent Vehicles*, (Under Review).

[TAES.*review*] [28] **Gyu Seon Kim**, Hyojun Ahn, Zhu Han, and Joongheon Kim, "Quantum Reinforcement Learning for Carbon-Aware Energy-Efficient Air Traffic Control," *IEEE Transactions on Aerospace and Electronic Systems*, (Under Review).

[ICTE.*review*] [27] **Gyu Seon Kim**, Soohyun Park, Soyi Jung, and Joongheon Kim, "Learning-based Aircraft Taxi Routing for Traffic Management: Empirical Evaluation Study for Hartsfield-Jackson Atlanta International Airport," *ICT Express (Elsevier)*, (Under Review).

◄ Revision ►

[TASE.*revision*] [26] **Gyu Seon Kim** and Joongheon Kim, "(Double-Blind Review)," *IEEE Transactions on Automation Science and Engineering*, (Under Revision).

[IoTJ.*revision*] [25] **Gyu Seon Kim**, Soohyun Park, and Joongheon Kim, "LLM-Guided Deep Reinforcement Learning for Robust and Autonomous Anti-Drone Systems," *IEEE Internet of Things Journal*, (Under Revision).

[TSC.*revision*] [24] **Gyu Seon Kim**, Soohyun Park, and Joongheon Kim, "Dynamic Sun Tracking Control for Efficient Solar Power Charging in Sustainable Cube-Satellites," *IEEE Transactions on Sustainable Computing*, (Under Revision).

[TMC.*revision*] [23] **Gyu Seon Kim**, Jaehyun Chung, Soyi Jung, Soohyun Park, Trung Q. Duong, and Joongheon Kim, "Joint Mobility Control and Networking in Autonomous Aircraft Cooperation: A Quantum Reinforcement Learning Approach," *IEEE Transactions on Mobile Computing*, (Under Revision).

◄ **Published** ►

[TON.*accept*] [22] Hyojun Ahn, Seungcheol Oh, **Gyu Seon Kim**, Soohyun Park, Soyi Jung, and Joongheon Kim, "Hallucination-Aware Hierarchical LLM for Autonomous UAV Mobility Control: A Safe Reinforcement Learning Approach," *IEEE Transactions on Networking*, 2026 (Early Access).

[IoTJ.*accept*] [21] **Gyu Seon Kim**, Emily Jimin Roh, Soohyun Park, and Joongheon Kim, "Exponentially Scalable Quantum Reinforcement Learning for Large-Scale Network Scheduling," *IEEE Internet of Things Journal*, 2026 (Early Access).

[TMC.*accept*] [20] **Gyu Seon Kim**, Emily Jimin Roh, Soyi Jung, Soohyun Park, and Joongheon Kim, "Scale-Reconfigurable Multi-Agent Reinforcement Learning for Aerial Networks in Dynamic Environments," *IEEE Transactions on Mobile Computing*, 2026 (Early Access).

[CM.*accept*] [19] **Gyu Seon Kim**, Jaehyun Chung, Soohyun Park, Soyi Jung, Trung Q. Duong, and Joongheon Kim, "Quantum Learning for Autonomous Aircraft Control: A Reinforcement Learning Perspective," *IEEE Communications Magazine*, 2026 (Early Access).

[NM.*accept*] [18] Hyunsoo Lee, **Gyu Seon Kim**, Emily Jimin Roh, Soyi Jung, Soohyun Park, Trung Q. Duong, and J.Kim, "Quantum Jump to Connected Worlds: Paradigm Shifts in Mobility and Network Applications via Quantum Neural Networks," *IEEE Network Magazine*, 2026 (Early Access).

[TIV.*accept*] [17] **Gyu Seon Kim**, Sungjoon Lee, Taejin Woo, and Soohyun Park, "Cooperative Reinforcement Learning for Military Drones over Large-Scale Battlefields," *IEEE Transactions on Intelligent Vehicles*, 2024 (Early Access).

◄ **2026** ►

[IoTJ.26.08] [16] **Gyu Seon Kim**, Chaemoon Im, Yeong Goo Kim, Jaekyoung Ha, and Joongheon Kim, "Learning-based Resilient Dynamic Routing for Fault-Tolerant Robust LEO Satellite Networks," *IEEE Internet of Things Journal*, vol. 13, no. 16, pp. 37402-37414, August 2026.

[TMC.26.07] [15] Soohyun Park, **Gyu Seon Kim**, Soyi Jung, Zhu Han, and Joongheon Kim, "Joint Sustainable Control and Quantum Reinforcement Learning for Energy-Efficient CubeSats," *IEEE Transactions on Mobile Computing*, vol. 25, no. 7, pp. 10385-10401, July 2026.

[TAES.26.03] [14] **Gyu Seon Kim**, Jaehyun Chung, Soyi Jung, Soohyun Park, and Joongheon Kim, "Quantum Reinforcement Learning for Joint Control, Communication, and Computing in Stabilized Reusable Space Rocket," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 62, pp. 3838-3863, March 2026.

[TAES.26.02] [13] **Gyu Seon Kim**, Soohyun Park, Soyi Jung, David Mohaisen, and Joongheon Kim, "Integrated Control, Communication, and Computing for Mission-Critical Embedded Unmanned Aerial Vehicles," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 62, pp. 682-703, February 2026.

[IoTJ.26.02] [12] Hyojun Ahn, **Gyu Seon Kim**, In-Sop Cho, Soyi Jung, and Joongheon Kim, "Hybrid Large Language Models and Reinforcement Learning for Energy-Efficient Multi-Satellite Scheduling: Boosting the Performance from Scratch," *IEEE Internet of Things Journal*, vol. 13, no. 3, pp. 5379-5392, February 2026.

[TMC.26.01] [11] **Gyu Seon Kim**, Yeryeong Cho, Jaehyun Chung, Soohyun Park, Soyi Jung, Zhu Han, and Joongheon Kim, "Quantum Multi-Agent Reinforcement Learning for Cooperative Mobile Access in Space-Air-Ground Integrated Networks," *IEEE Transactions on Mobile Computing*, vol. 25, no. 1, pp. 1200-1218, January 2026.

◄ **2025** ►

[TON.25.08] [10] Hankyul Baek†, **Gyu Seon Kim (co-first)**†, Soohyun Park, Andreas F. Molisch, and Joongheon Kim, "Slimmable Federated Reinforcement Learning for Energy-Efficient Proactive Caching," *IEEE Transactions on Networking*, vol. 33, no. 4, pp. 2079-2094, August 2025.
†These authors contributed equally.

[TAES.25.08] [9] **Gyu Seon Kim**, Yeryeong Cho, Soohyun Park, Soyi Jung, and Joongheon Kim, "Quantum Multiagent Reinforcement Learning for Joint Cube Satellites and High-Altitude Long-Endurance Aerial Vehicles in SAGIN," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 61, no. 4, pp. 9490-9510, August 2025.

[IoTJ.25.07] [8] **Gyu Seon Kim**, Sungjoon Lee, In-Sop Cho, Soohyun Park, and Joongheon Kim, "Quantum Reinforcement Learning for Lightweight LEO Satellite Routing," *IEEE Internet of Things Journal*, vol. 12, no. 14, pp. 28986-29004, July 2025.

[IoTJ.25.07] [7] Soohyun Park, **Gyu Seon Kim**, and Joongheon Kim, "Joint Quantum Reinforcement Learning and Neural Myerson Auction for High-Quality Digital-Twin Services in Multi-Tier Networks," *IEEE Internet of Things Journal*, vol. 12, no. 13, pp. 23722-23735, July 2025.

◄ **2024** ►

[IoTJ'24.12] [6] Soohyun Park†, **Gyu Seon Kim (co-first)**†, Soyi Jung, and Joongheon Kim, "Markov Decision Policies for Distributed Angular Routing in LEO Mobile Satellite Constellation Networks," *IEEE Internet of Things Journal*, vol. 11, no. 23, pp. 38744-38754, December 2024.
†These authors contributed equally.

[CM'24.10] [5] Soohyun Park, **Gyu Seon Kim**, Zhu Han, and Joongheon Kim, "Quantum Multi-Agent Reinforcement Learning is All You Need: Coordinated Global Access in Integrated TN/NTN Cube-Satellite Networks," *IEEE Communications Magazine*, vol. 42, no. 10, pp. 86-92, October 2024.

[TVT'24.08] [4] **Gyu Seon Kim**, Jaehyun Chung, and Soohyun Park, "Realizing Stabilized Landing for Computation-Limited Reusable Rockets: A Quantum Reinforcement Learning Approach," *IEEE Transactions on Vehicular Technology*, vol. 73, no. 8, pp. 12252-12257, August 2024.

[Access'24.08] [3] Hyunsoo Lee, **Gyu Seon Kim**, Minseok Choi, Hankyul Baek, and Soohyun Park, "Farthest Agent Selection with Episode-Wise Observations for Real-Time Multi-Agent Reinforcement Learning Applications," *IEEE Access*, vol. 12, pp. 108504-108514, August 2024.

◂ 2023 ▸

[ETRI'23.10] [2] **Gyu Seon Kim**, Haemin Lee, Soohyun Park, and Joongheon Kim, "Joint Frame Rate Adaptation and Object Recognition Model Selection for Stabilized Unmanned Aerial Vehicle Surveillance," *ETRI Journal*, vol. 45, no. 5, pp. 811-821, October 2023.

[TIV'23.08] [1] Chanyoung Park, **Gyu Seon Kim**, Soohyun Park Soyi Jung, and Joongheon Kim, "Multi-Agent Reinforcement Learning for Cooperative Air Transportation Services in City-Wide Autonomous Urban Air Mobility," *IEEE Transactions on Intelligent Vehicles*, vol. 8, no. 8, pp. 4016-4030, August 2023.

Conferences

◂ Top-Tiers ▸

[GLOBEC'25] [12] Hyojun Ahn, Seungcheol Oh, **Gyu Seon Kim**, Soyi Jung, Soohyun Park, and Joongheon Kim, "Hallucination-Aware Generative Pretrained Transformer for Cooperative Aerial Mobility Control," in *Proc. IEEE Global Communications Conference* (*GLOBECOM*), Taipei, Taiwan, December 2025, pp. 5369-5374.

[WiOpt'25] [11] **Gyu Seon Kim**, Jaehyun Chung, Trung Q. Duong, Soohyun Park, and Joongheon Kim, "Stabilized Robust Control for Lightweight Autonomous Aircraft Mobility: A Quantum Reinforcement Learning Approach," in *Proc. IEEE/IFIP International Symposium on Modeling and Optimization in Mobile, Ad hoc, and Wireless Networks* (*WiOpt*), Linköping, Sweden, May 2025, pp. 1-8.

[ICASSP'25] [10] **Gyu Seon Kim**, Samuel Yen-Chi Chen, Soohyun Park, and Joongheon Kim, "Quantum Reinforcement Learning for Coordinated Satellite Systems", in *Proc. IEEE International Conference on Acoustics, Speech, and Signal Processing* (*ICASSP*), Hyderabad, India, April 2025, pp. 1-5.

[WiOpt'24] [9] Sungjoon Lee, **Gyu Seon Kim**, Soohyun Park, and Joongheon Kim, "Advanced Taxiing Path Guidance using Multi-Agent Reinforcement Learning for Air Traffic Management," in *Proc. IEEE/IFIP International Symposium on Modeling and Optimization in Mobile, Ad hoc, and Wireless Networks* (*WiOpt*), Seoul, Republic of Korea, October 2024, pp. 305-312.

[ICC'23] [8] Chanyoung Park, Soohyun Park, **Gyu Seon Kim**, Soyi Jung, Jae-Hyun Kim, and Joongheon Kim, "Multi-Agent Deep Reinforcement Learning for Efficient Passenger Delivery in Urban Air Mobility," in *Proc. IEEE International Conference on Communications* (*ICC*), Rome, Italy, June 2023, pp. 5689-5694.

◂ Others ▸

[PQAI'25] [7] **Gyu Seon Kim**, Samuel Yen-Chi Chen, Soohyun Park, and Joongheon Kim, "Satellite Constellation Scheduling with Quantum Reinforcement Learning", in *Proc. Springer Nature, Frontiers in Industrial Engineering. Purdue Quantum AI* (*Communications in Computer and Information Science, vol. 2724*), West Lafayette, IN, USA, October 2025, pp. 51-56.

[ICTC'25] [6] **Gyu Seon Kim**, In-Sop Cho, and Joongheon Kim, "Strategic Role of LEO Satellites in the 6G Roadmap and Orbital Analysis," in *Proc. IEEE International Conference on Information and Communication Technology Convergence* (*ICTC*), Jeju, Republic of Korea, October 2025, pp. 1996-2001.

[ICTC'24] [5] Hyunwoo Nam, **Gyu Seon Kim**, and Joongheon Kim, "Bicycle Rental Predictive Modeling with Machine Learning for Carbon Neutrality," in *Proc. IEEE International Conference on Information and Communication Technology Convergence* (*ICTC*), Jeju, Republic of Korea, October 2024, pp. 2033-2038.

[ICTC'24] [4] **Gyu Seon Kim**, Soohyun Park, and Joongheon Kim, "Quantum Reinforcement Learning: An Overview," in *Proc. IEEE International Conference on Information and Communication Technology Convergence* (*ICTC*), Jeju, Republic of Korea, October 2024, pp. 1145-1150.

[APWCS'24] [3] Soohyun Park, **Gyu Seon Kim**, Soyi Jung, and Joongheon Kim, "Quantum Multi-Agent Reinforcement Learning Software Design and Visual Simulations for Multi-Drone Mobility Control," in *Proc. IEEE VTS Asia Pacific Wireless Communications Symposium* (*APWCS*), Singapore, August 2024, pp. 1-5.

[IJCAI'24] [2] **Gyu Seon Kim**, Yeryeong Cho, Soohyun Park, Soyi Jung, and Joongheon Kim, "Realizing Cooperative Global Internet Services for Space-Air-Ground Integrated Networks: A Quantum Multi-Agent Reinforcement Learning Framework," *International Joint Conference on Artificial Intelligence* (*IJCAI*) *Workshop on Quantum Algorithms, Optimization, and AI*, Jeju, Republic of Korea, August 2024.

[ICOIN'23] [1] **Gyu Seon Kim**, Hyunsoo Lee, Chanyoung Park, Soyi Jung, Jae-Hyun Kim and Joongheon Kim, "Hybrid MAC for Military UAV Networks," in *Proc. IEEE International Conference on Information Networking* (*ICOIN*), Bangkok, Thailand, January 2023, pp. 656-659.

Publications (Domestic)

Journals (KCI)

◂ Revision ▸

[KIISE.*review*] [5] **Gyu Seon Kim**, Soohyun Park, and Joongheon Kim, "Reinforcement Learning-based LEO Satellite Control and Communication," *Journal of Korean Institute of Information Scientists and Engineers* (*KIISE*), (Under Revision).

◂ 2026 ▸

[KIISE’26.02] [4] Sungjoon Lee, **Gyu Seon Kim**, Taejin Woo, and Soohyun Park, ”Design of a Deep Reinforcement Learning Algorithm for Spatially Adaptive UAV Autonomous Navigation based on Transfer Learning,” *Journal of Korean Institute of Information Scientists and Engineers* (*KIISE*), vol. 53, no. 2, pp. 101-108, February 2026.

◂ 2024 ▸

[KICS’24.03] [3] Jaehyun Chung, **Gyu Seon Kim**, Soohyun Park, Joongheon Kim, and W. Yun, ”Reinforcement Learning-Based Counter Measure Tactics to Avoid Torpedo Threats,” *Journal of Korean Institute of Communications and Information Sciences* (*KICS*), vol. 49, no. 3, pp. 333-345, March 2024.

◂ 2023 ▸

[KICS’23.09] [2] **Gyu Seon Kim**, Soohyun Park, Joongheon Kim, Yeong Goo Kim, Jaekyoung Ha, and Byung Hyun Jun, ”Reinforcement Learning-Based Dynamic Routing for Robust Optimization of Low Earth Orbit (LEO) Satellite Communication Networks,” *Journal of Korean Institute of Communications and Information Sciences* (*KICS*), vol. 48, no. 9, pp. 1123-1134, September 2023.

[KICS’23.02] [1] Chanyoung Park, **Gyu Seon Kim**, Kyeongjin Lee, and Ilsoo Yun, ”Multi-Agent Deep Reinforcement Learning for Efficient Unattended Information Gathering and Monitoring of Autonomous UAM Systems,” *Journal of Korean Institute of Communications and Information Sciences* (*KICS*), vol. 48, no. 2, pp. 176-184, February 2023.

Conferences

◂ 2026 ▸

[KICS’26] [17] **Gyu Seon Kim** and Joongheon Kim, ”LLM-based Reinforcement Learning for UAV Control in Defense Systems,” in *Proc. Korean Institute of Communications and Information Sciences* (*KICS*) *Conference*, Yongpyeong, Republic of Korea, February 2026, pp. 534-535.

◂ 2025 ▸

[IEIE’25] [16] **Gyu Seon Kim** and Joongheon Kim, ”Lightweight Control Modeling via Quantum Reinforcement Learning,” in *Proc. Institute of Electronics and Information Engineers* (*IEIE*) *Conference*, Gonjiam, Republic of Korea, November 2025, pp. 1584-1587.

[KICS’25] [15] Jaehyun Chung, Hyojun Ahn, **Gyu Seon Kim**, and Joongheon Kim, ”Recent Trends in Transformer Optimization for Improved Energy Efficiency,” in *Proc. Korean Institute of Communications and Information Sciences* (*KICS*) *Conference*, Jeju, Republic of Korea, June 2025, pp. 574-575.

[JCCI’25] [14] **Gyu Seon Kim**, Jaeyoung Choe, Soohyun Park, and Joongheon Kim, ”Federated Reinforcement Learning-based Caching using Slimmable Neural Networks,” *Joint Conference on Communications and Information* (*JCCI*), Gangneung, Republic of Korea, April 2025.

[KICS’25] [13] **Gyu Seon Kim**, Sungjoon Lee, Taejin Woo, Soohyun Park, and Joongheon Kim, ”Reinforcement Learning-based Military UAV Control,” in *Proc. Korean Institute of Communications and Information Sciences* (*KICS*) *Conference*, Yongpyeong, Republic of Korea, February 2025, pp. 291-292.

[KICS’25] [12] Yun Seong Na, **Gyu Seon Kim**, and Joongheon Kim, ”Optimal Algorithm Proposal for Urban Road Pathfinding Using Reinforcement Learning,” in *Proc. Korean Institute of Communications and Information Sciences* (*KICS*) *Conference*, Yongpyeong, Republic of Korea, February 2025, pp. 520-521.

◂ 2024 ▸

[KICS’24] [11] **Gyu Seon Kim**, Soohyun Park, and Joongheon Kim, ”Reinforcement Learning-based Self-Driving Vehicle Control for Carbon Neutrality,” in *Proc. Korean Institute of Communications and Information Sciences* (*KICS*) *Conference*, Gyeongju, Republic of Korea, November 2024, pp. 207-208.

[KICS’24] [10] **Gyu Seon Kim**, Soohyun Park, and Joongheon Kim, ”Reinforcement Learning-based Aircraft Longitudinal Static Stability Control,” in *Proc. Korean Institute of Communications and Information Sciences* (*KICS*) *Conference*, Jeju, Republic of Korea, June 2024, pp. 1,094-1,095.

[KICS’24] [9] **Gyu Seon Kim**, Hankyul Baek, Soohyun Park, and Joongheon Kim, ”Reinforcement Learning-based Lunar Module Landing Controller Design,” in *Proc. Korean Institute of Communications and Information Sciences* (*KICS*) *Conference*, Yongpyeong, Republic of Korea, January 2024, pp. 134-135.

[KICS’24] [8] **Gyu Seon Kim**, Hankyul Baek, Soohyun Park, and Joongheon Kim, ”Slimmable Neural Networks Reinforcement Learning-based Robust Low Earth Orbit (LEO) Satellite Routing,” in *Proc. Korean Institute of Communications and Information Sciences* (*KICS*) *Conference*, Yongpyeong, Republic of Korea, January 2024, pp. 76-77.

◂ 2023 ▸

[KAIC’23] [7] Emily Jimin Roh, **Gyu Seon Kim**, Hankyul Baek, Soohyun Park, Soyi Jung, and Joongheon Kim, ”Autonomous Torpedo Maneuvering and Waypoint Optimization via Reinforcement Learning,” in *Proc. Korea Artificial Intelligence Conference* (*KAIC*), Jeju, Republic of Korea, September 2023, pp. 177-178.

[KICS’23] [6] **Gyu Seon Kim**, Soohyun Park, and Joongheon Kim, ”Utilizing Reinforcement Learning for Enhanced Efficiency in Fighter Airport Scheduling: A Case Study of Osan Air Base,” in *Proc. Korean Institute of Communications and Information Sciences* (*KICS*) *Conference*, Jeju, Republic of Korea, June 2023, pp. 867-868.

[JCCI’23] [5] **Gyu Seon Kim**, Soohyun Park, and Joongheon Kim, ”Low-Earth Orbit Satellite Routing based on Reinforcement Learning,” *Joint Conference on Communications and Information* (*JCCI*), Yeosu, Republic of Korea, May 2023.

[JCCI’23] [4] **Gyu Seon Kim**, Soohyun Park, and Joongheon Kim, ”A Mathematical Approach to Reinforcement Learning through the Bellman Equation,” *Joint Conference on Communications and Information* (*JCCI*), Yeosu, Republic of Korea, May 2023.

[KICS'23] [3] Hyunsoo Lee, **Gyu Seon Kim** and Joongheon Kim, "Trends in Optimal Routing Technologies for Low Earth Orbit Satellite," in *Proc. Korean Institute of Communications and Information Sciences (KICS) Conference*, Yongpyeong, Republic of Korea, February 2023, pp. 406-407.

[KICS'23] [2] **Gyu Seon Kim** and Joongheon Kim, "Communication & Sun-Pointing Mode Control Model For Cube Satellite," in *Proc. Korean Institute of Communications and Information Sciences (KICS) Conference*, Yongpyeong, Republic of Korea, February 2023, pp. 1088-1089.

◂ 2022 ▸

[KICS'22] [1] **Gyu Seon Kim**, Hyunsoo Lee, Soyi Jung, J. H. Kim, and Joongheon Kim, "Self-Learning-Based Hybrid MAC for Military UAV Networks," in *Proc. Korean Institute of Communications and Information Sciences (KICS) Conference*, Gyeongju, Republic of Korea, November 2022, pp. 13-14.

Patents

United States

◂ 2025 ▸

[U.S.'25.12] [9] Patent Application: No. 19/425,505, "Method and Apparatus for Routing Optimal Path in Low Earth Orbit Constellation Satellite Network System," December 2025.

Republic of Korea

◂ 2026 ▸

[KR'26.07] [8] Patent Application: No. 10-2026-0133424, "Autonomous Unmanned Aerial Vehicle Mobility Control System and Method using Hierarchical Large Language Models and Safe Reinforcement Learning," July 2026.

[KR'26.07] [7] Patent Application: No.10-2026-0128332, "Energy-Efficient Multi-Satellite Scheduling System and Method for Satellite Network using Initial Policy based on Large-Scale Language Model for Low-Earth Orbit Satellite Cluster," July 2026.

◂ 2025 ▸

[KR'25.08] [6] Patent Application: No. 10-2025-0111715, "Multi-Agent Reinforcement Learning Based Anti-Drone Control System and Method," August 2025.

- ∗ Patent Registration: No. 10-2991143, "Multi-Agent Reinforcement Learning Based Anti-Drone Control System and Method," July 2026. (Ministry of Intellectual Property, Republic of Korea)

[KR'25.01] [5] Patent Application: No. 10-2025-0004523, "Multi-Agent Reinforcement Learning Device and Method," January 2025.

◂ 2024 ▸

[KR'24.12] [4] Patent Application: No. 10-2024-0191045, "Ground Station and Method for Scheduling Non-Terrestrial Network Devices for Global Access in a Space-Air-Ground Integrated Network System," December 2024.

[KR'24.12] [3] Patent Application: No. 10-2024-0199342, "Ground Station and Method for Cube Satellite Constellation Scheduling Based on Quantum Multi-Agent Reinforcement Learning," December 2024.

[KR'24.12] [2] Patent Application: No. 10-2024-0199295, "Unmanned Aerial Vehicle Device Performing Communication in a Multi-Military Unmanned Aerial Vehicle Network System and Communication Method of the Unmanned Aerial Vehicle Device," December 2024.

[KR'24.12] [1] Patent Application: No. 10-2024-0191045, "Method and Apparatus For Routing Optimal Path in Low Earth Orbit Constellation Satellite Network System," December 2024.

Review Experience

- **IEEE Transactions on Aerospace and Electronic Systems**
- **IEEE Transactions on Vehicular Technology**
- **IEEE Transactions on Mobile Computing**
- **IEEE Transactions on Communications**
- **IEEE Transactions on Industrial Informatics**
- **IEEE Transactions on Network and Service Management**
- **IEEE Transactions on Cognitive Communications and Networking**
- **IEEE Internet of Things Journal**
- **IEEE Communications Magazine**
- **IEEE Access**
- **IEEE/KICS Journal of Communications and Networks**
- **IEEE/KICS ICT Express**
- **Springer Nature Aerospace Systems**
- **ACM Computing Surveys**
- **KSII Transactions on Internet and Information Systems**
- **ETRI Journal**

Talks

- **9. Workshop on Implementing Autonomous Vehicles using Modern Reinforcement Learning (01/2026, Seoul, Republic of Korea)**
  – *Implementation of Multi-Agent Reinforcement Learning-based Unmanned Vehicle Control*
- **8. Quantum Security Research Society Mid-Year Workshop (07/2025, Sejong, Republic of Korea)**
  – *Theory and Applications of Quantum Reinforcement Learning*
- **7. Understanding and Practice of Artificial Intelligence for Smart Mobility (03/2025, Seoul, Republic of Korea)**
  – *Multi-Agent Reinforcement Learning-based Autonomous Urban Air Mobility Control for Aerial Transportation Services*
  – *Multi-Agent Reinforcement Learning-based Advanced Air Traffic Management & Control*
- **6. Workshop on Implementing Autonomous Vehicles using Modern Reinforcement Learning (01/2025, Seoul, Republic of Korea)**
  – *Implementation of Multi-Agent Reinforcement Learning-based Unmanned Vehicle Control*
- **5. A3 Foresight Program (06/2024, Jeju, Republic of Korea)**
  – *Quantum Multi-Agent Reinforcement Learning for Joint Cube-Satellites and High-Altitude Long-Endurance Aerial Networks in SAGIN*
- **4. Implementation and Simulation of Reinforcement Learning Algorithms Based on Unity (02/2024, Seoul, Republic of Korea)**
  – *Application of Reinforcement Learning in Unity Environment Using ML-Agents – Practice in Various Scenarios*
- **3. A3 Foresight Program (02/2023, Seoul, Republic of Korea)**
  – *Communication & Sun-pointing Mode Control Model for Cube Satellite*
- **2. Implementation and Simulation of Unity-Based Reinforcement Learning Algorithms for Telecommunication Networks (02/2023, Seoul, Republic of Korea)**
  – *Application of Reinforcement Learning in Unity Environment Using ML-Agents – Practice in Various Scenarios*
- **1. A3 Foresight Program (12/2022, Tokyo, Japan)**
  – *Hybrid MAC for Military UAV Networks*

Teaching Assistants

- **SW Programming Basics (GECT002), Dept. of ECE, Korea University**
  – *Spring 2025*

References

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